

A presentation given on 22-May-2024 at an event hosted by the Institute of Physics and Engineering in Medicine (London, UK).

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Racial bias in pulse oximetry: next steps

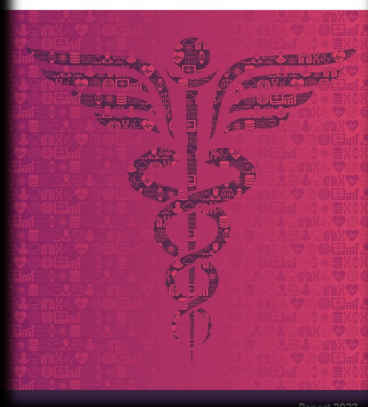
Dr Peter H. Charlton

University of Cambridge
City, University of London



New England Journal of Medicine
Racial bias in pulse oximetry
measurement

Equity in Medical Devices:
Independent Review



A group of people are seated at a long wooden table in what appears to be a meeting or workshop. In the foreground, a person is writing in a yellow notebook with a pen. Other people are visible in the background, some looking at documents or devices. The scene is brightly lit, possibly by natural light from a window.

Resources: Wealth of expertise
across disciplines

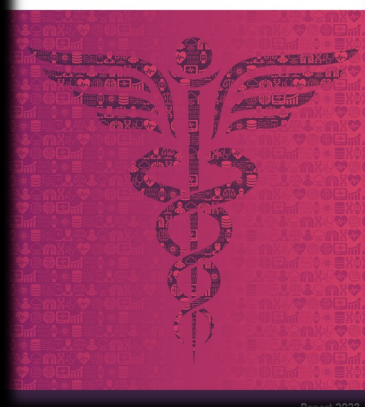
Infrastructure: NHS ecosystem for
research and implementation

Timeliness: Recent report and
increasing use of pulse oximetry



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Guidance

Regulation

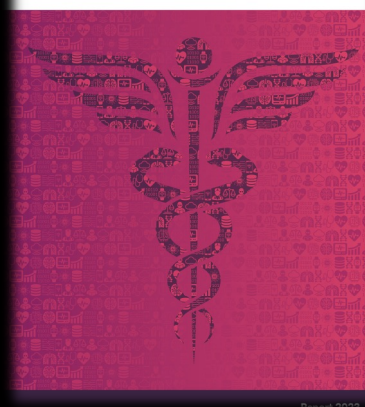
Device
development

Research



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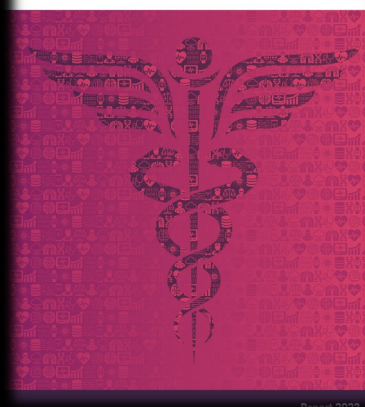
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Guidance

Recommendation 1:

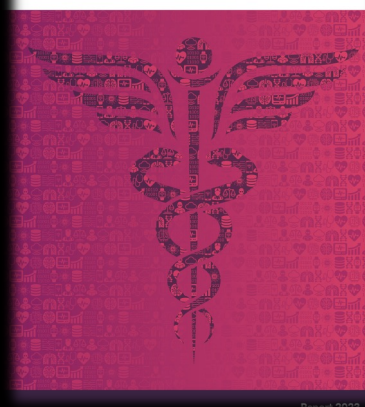
“Regulators, developers, manufacturers and healthcare professionals should take immediate mitigation actions to ensure existing pulse oximeter devices in the NHS can be used safely and equitably for all patient groups across the range of skin tones.”

- Guidance for use
- Looking at changes in readings
- Using trends rather than thresholds to inform decisions
- Educate clinicians on pulse oximeter technology
- Manufacturers to update instructions for use
- Developers and manufacturers to report performance across subgroups



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Recommendation 2:

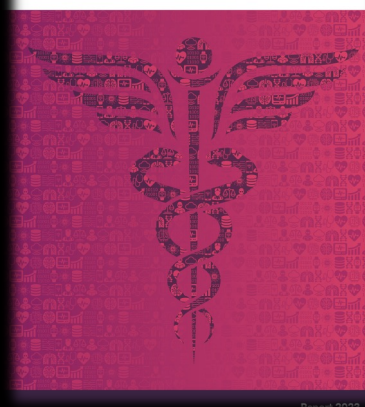
“The MHRA and approved bodies for medical devices should strengthen the standards for approval of new pulse oximeter devices to include sufficient clinical data to demonstrate accuracy overall and in groups with darker skin tones. Greater population representativeness in testing and calibration of devices should be stipulated.”

- Ensure diversity in validity data:
 - no. participants, range of skin tones, range of oxygenation levels
- Validate in intended use population and setting
- Comply with ISO standard
- Make healthcare equity impact assessments an essential requirement



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Recommendation 3:

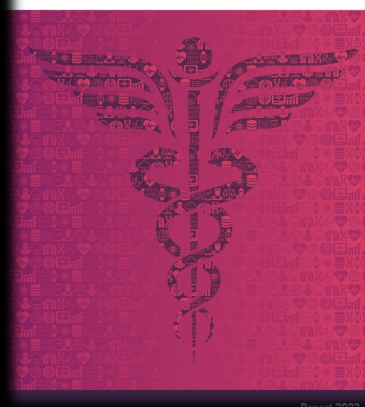
“Innovators, researchers and manufacturers should cooperate with public and patient participants to design better, smarter oximeters using innovative technologies to produce devices that are not biased by skin tone.”

- Enhanced algorithms to address measurement bias
- Multi-wavelength systems which measure and correct for skin pigmentation



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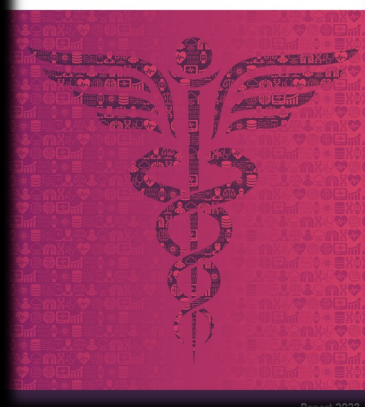
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Recommendation 1:

“Regulators, developers, manufacturers and healthcare professionals should take immediate mitigation actions to ensure existing pulse oximeter devices in the NHS can be used safely and equitably for all patient groups across the range of skin tones.”

- Guidance for use
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- **Using trends rather than thresholds to inform decisions**
- Educate clinicians on pulse oximeter technology
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Using trends rather than thresholds to inform decisions

Chart 1: The NEWS scoring system

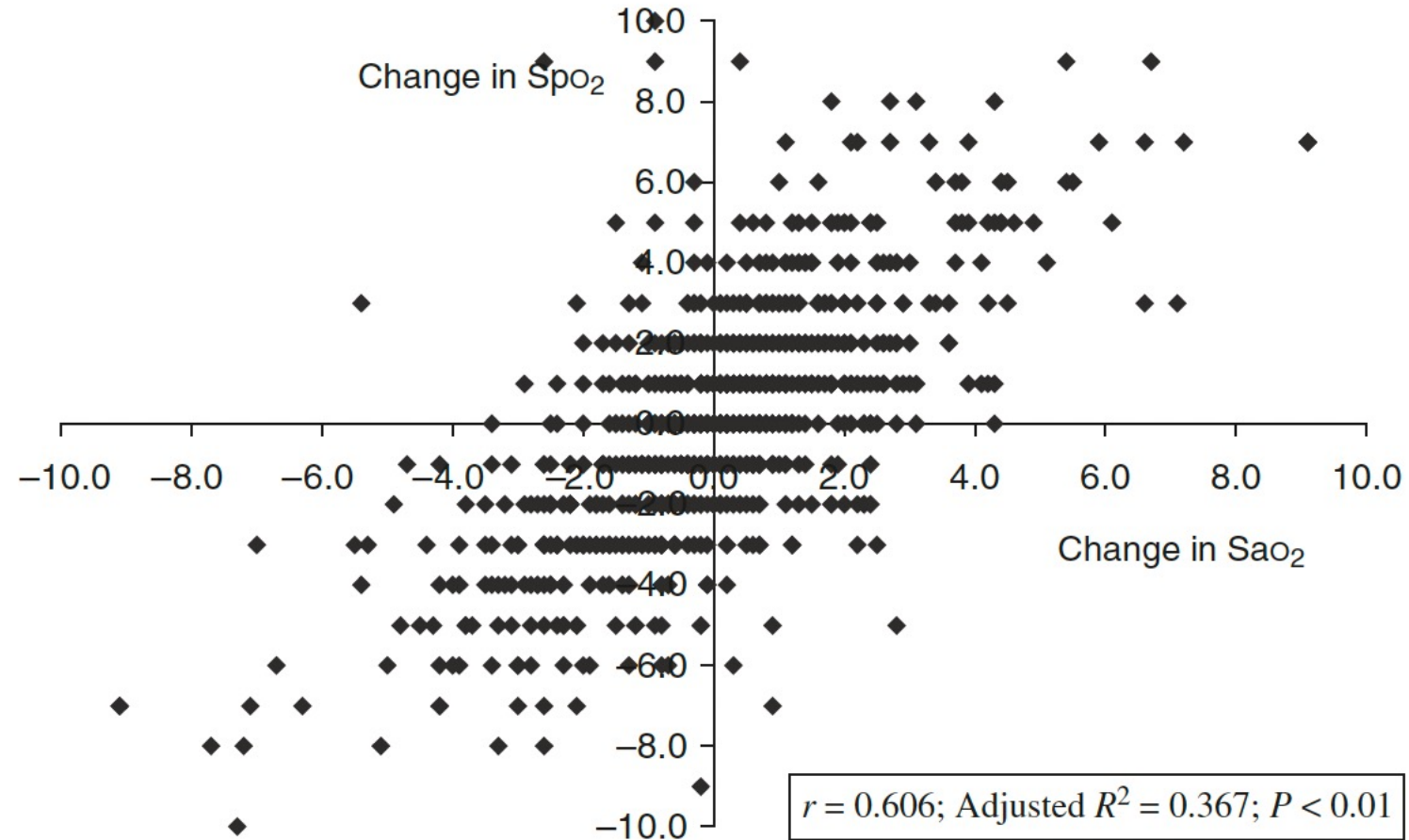
Physiological parameter	3	2	1	Score 0	1	2	3
Respiration rate (per minute)	≤8		9–11	12–20		21–24	≥25
SpO ₂ Scale 1 (%)	≤91	92–93	94–95	≥96			
SpO ₂ Scale 2 (%)	≤83	84–85	86–87	88–92 ≥93 on air	93–94 on oxygen	95–96 on oxygen	≥97 on oxygen
Air or oxygen?		Oxygen		Air			
Systolic blood pressure (mmHg)	≤90	91–100	101–110	111–219			≥220
Pulse (per minute)	≤40		41–50	51–90	91–110	111–130	≥131
Consciousness				Alert			CVPU
Temperature (°C)	≤35.0		35.1–36.0	36.1–38.0	38.1–39.0	≥39.1	

Using trends rather than thresholds to inform decisions

What if you don't have a baseline for the trend? *e.g.*

- First measurement
- Patient is sick at first encounter

What if pulse oximeters don't monitor trends accurately?



Using trends rather than thresholds to inform decisions

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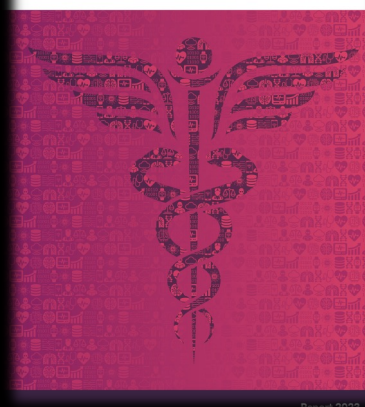
What if pulse oximeters don't monitor trends accurately?

What if the accuracy of trend monitoring differs with skin pigmentation?



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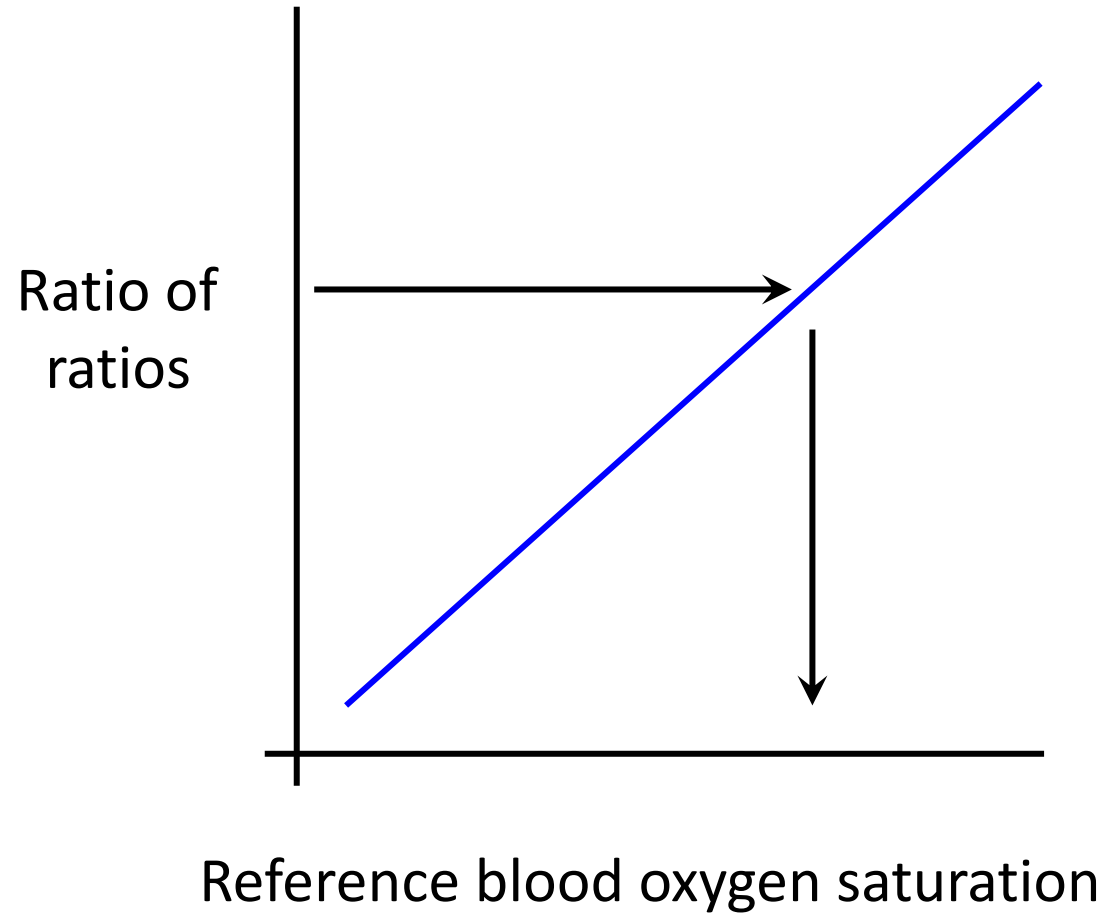
Device development

Recommendation 3:

“Innovators, researchers and manufacturers should cooperate with public and patient participants to design better, smarter oximeters using innovative technologies to produce devices that are not biased by skin tone.”

- **Enhanced algorithms to address measurement bias**
- **Multi-wavelength systems which measure and correct for skin pigmentation**

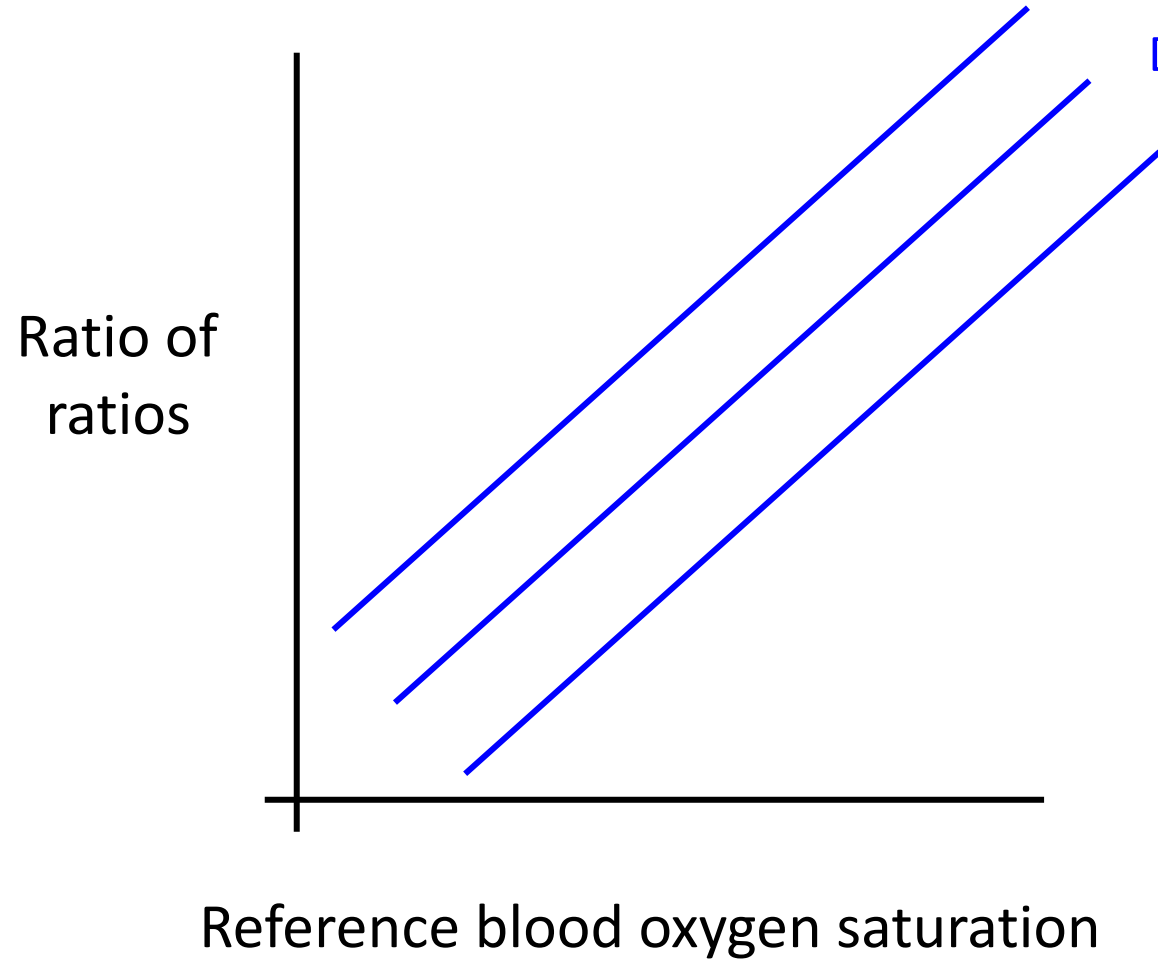
Pulse oximeter algorithms



Suggestion 1:

- Pulse oximeters are inaccurate because their equations are derived from measurements on predominantly White individuals.
- Increasing the diversity of training data would remove bias

Pulse oximeter algorithms



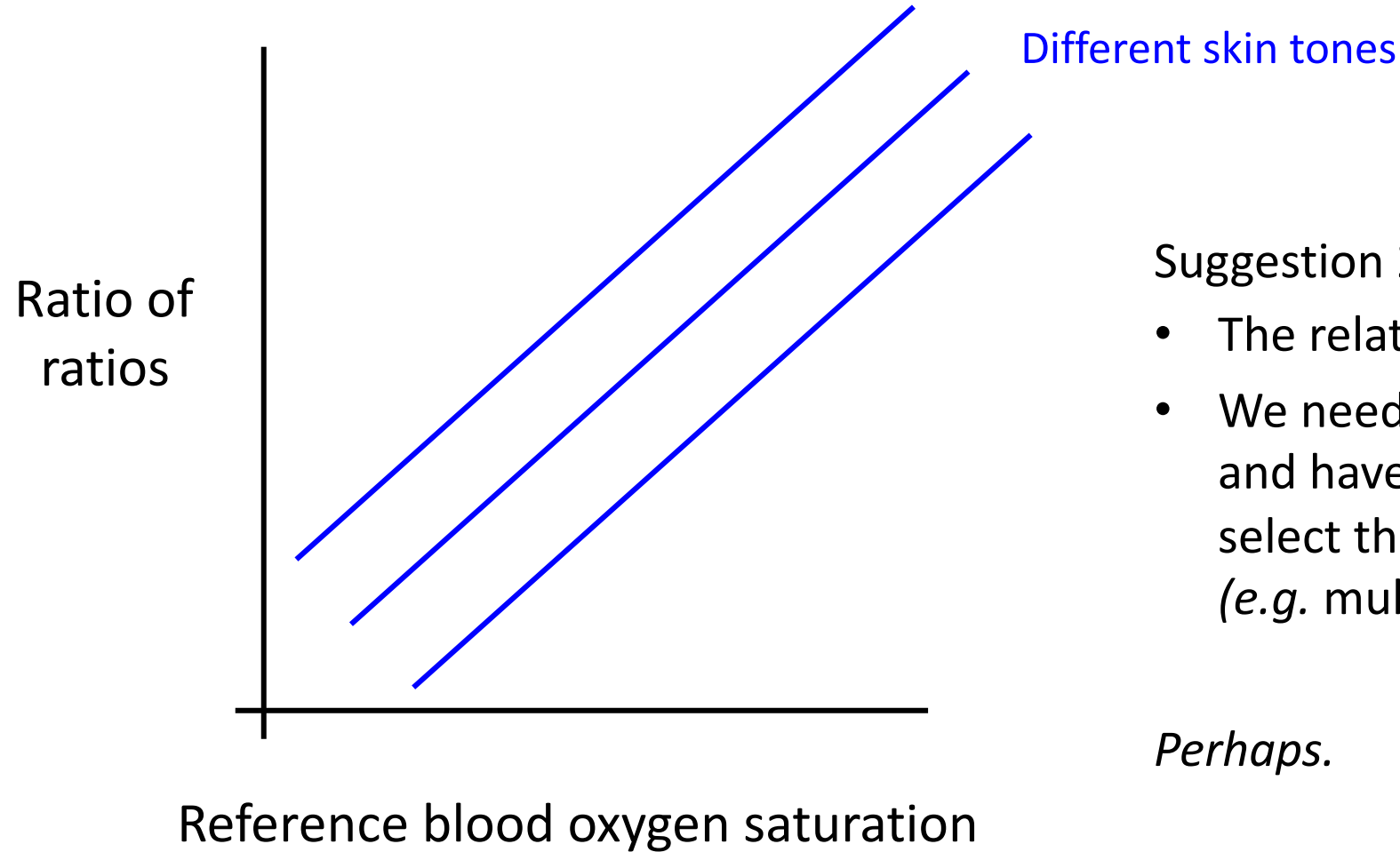
Different skin tones

Suggestion 1:

- Pulse oximeters are inaccurate because their equations are derived from measurements on predominantly White individuals.
- Increasing the diversity of training data would remove bias

It might reduce bias, but wouldn't eliminate it if the true relationship differs with skin tone.

Pulse oximeter algorithms

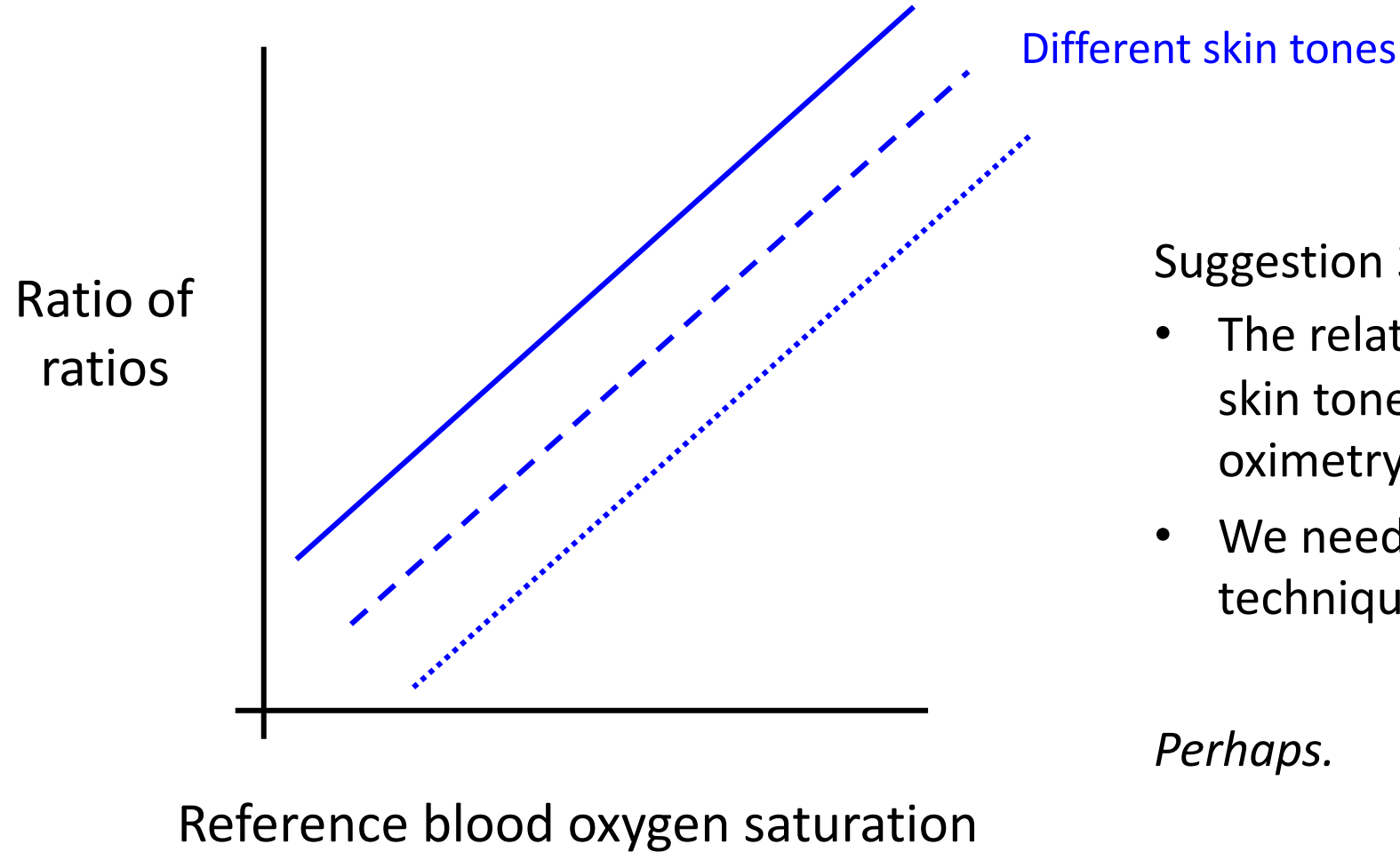


Suggestion 2:

- The relationships differ between skin tones.
- We need to learn the different relationships, and have a way of identifying skin tone to select the appropriate relationship to use (e.g. multiwavelength PPG).

Perhaps.

Pulse oximeter algorithms



Suggestion 3:

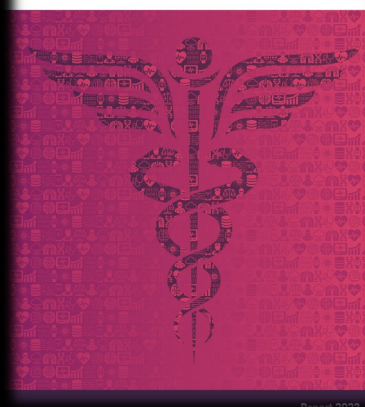
- The relationship(s) are less strong for some skin tones when using traditional pulse oximetry sensing.
- We need to identify improved sensing techniques (e.g. multiwavelength PPG).

Perhaps.



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Potential research directions

- Understanding current practice in the NHS
 - What pulse oximeters are being used?
 - How do they inform decision making?
- Level of inaccuracy in pulse oximeters in UK
 - Retrospective study using routine blood gas measurements as reference.
 - Consider not only differences in accuracy between skin tones, but also overall accuracy and precision.

Estimates of precision vary between studies, methodologies, and devices.

e.g. Limits of agreement of +/- 4%

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 - Consider differences between devices

“One oximeter overestimated SaO₂ (Philips, +0.9%), whereas the 3 others underestimated SaO₂ (Nonin −3.1%, Nellcor −0.3%, Masimo −0.2%).”

Potential research directions

- Understanding current practice in the NHS
 - What pulse oximeters are being used?
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- Level of inaccuracy in pulse oximeters in UK
 - Retrospective study using routine blood gas measurements as reference.
 - Consider not only differences in accuracy between skin tones, but also overall accuracy and precision.
 - Consider differences between devices
 - Consider additional populations (e.g. neonates)
- Impact of inaccuracies in pulse oximetry on outcomes
 - Assessing the scale of the problem in the UK
 - Retrospective study (e.g. using ICNARC) on occult hypoxaemia, or morbidity and mortality
 - Prospective study using objective quantification of skin tone
- Mitigation of differential bias with existing devices
 - Learn appropriate corrections
 - (somehow) apply these in practice (e.g. based on self-reported ethnicity)



Much work has been done to recognise the problem of racial bias in pulse oximetry.

There is much work to be done to mitigate against and overcome this problem. Areas for interdisciplinary work include:

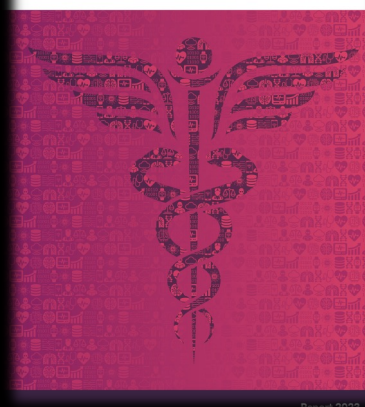
- Providing guidance
- Enhancing regulation
- Device development
- Research

Today we have a valuable opportunity to catalyse such work.



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